Integrating Powers of x

Exam code: 9MA0

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Written by: Paul

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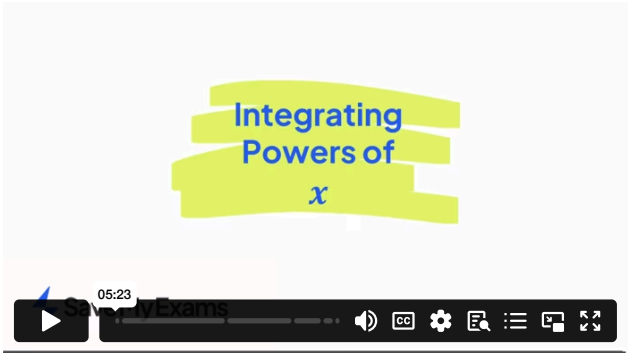
Help me understand



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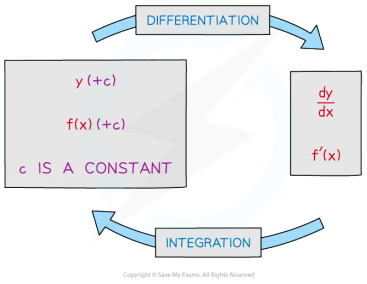
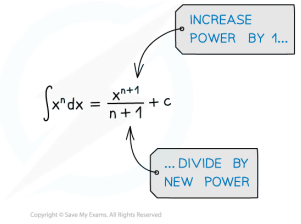
Integrating powers of x

What is integration?

- Integration is the **inverse** ('opposite') of differentiation



How do I integrate powers of x?

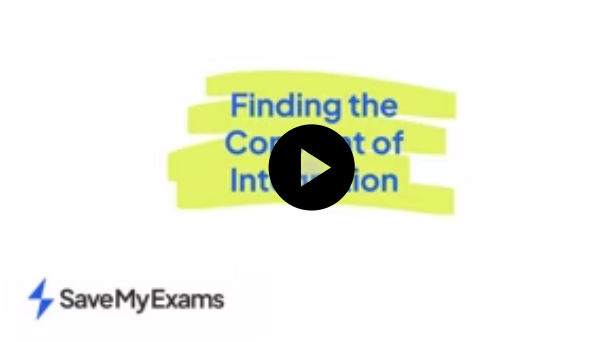


Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- This does not apply when the original power is -1
  - the new power would be 0 and division by 0 is undefined



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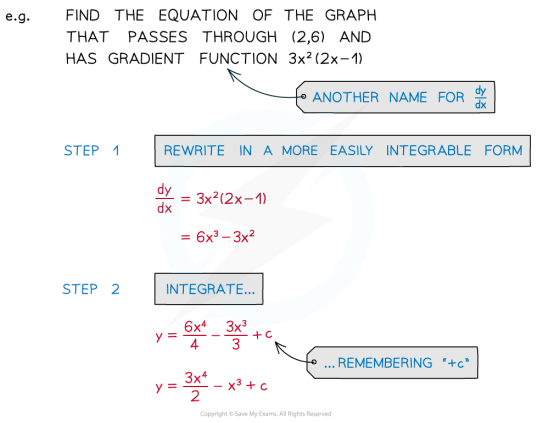
Finding the constant of integration

How do I find the constant of integration?

- STEP 1 Rewrite the function into a more easily **integrable** form
  - Each term needs to be a power of **x** (or a constant)
- STEP 2 Integrate each term and remember "+c"
  - Increase power by 1 and divide by new power
- STEP 3 Substitute the coordinates of a given point in to form an equation in **c**
  - Solve the equation to find **c**



$c = -10$

$\therefore \quad y = \frac{3}{2}x^4 - x^3 - 10$

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Worked Example

Given $f'(x) = \dfrac{(x+3)^2}{\sqrt{x}}$ and $f(1) = 25$,

find $f(x)$.

STEP 1

$$f'(x) = \frac{x^2 + 6x + 9}{x^{\frac{1}{2}}}$$

REWRITE IN A MORE EASILY INTEGRABLE FORM

$$f'(x) = \frac{x^2}{x^{\frac{1}{2}}} + \frac{6x}{x^{\frac{1}{2}}} + \frac{9}{x^{\frac{1}{2}}}$$

$$f'(x) = x^{\frac{3}{2}} + 6x^{\frac{1}{2}} + 9x^{-\frac{1}{2}}$$

STEP 2

$$f(x) = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{6x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{9x^{\frac{1}{2}}}{\frac{1}{2}} + c$$

INTEGRATE EACH TERM

REMEMBER "+c"!

$$f(x) = \frac{2}{5}x^{\frac{5}{2}} + 4x^{\frac{3}{2}} + 18x^{\frac{1}{2}} + c$$

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